

Shyam Harimohan Menon

Center for Computational Astrophysics
Flatiron Institute
NY 10010, USA

✉ smenon@flatironinstitute.org

🌐 <https://shm-1996.github.io>

🔗 [shm-1996](#)

🆔 [0000-0001-5944-291X](#)

Employment

- Sep. 2023 – **Flatiron Research Fellow (Joint Rutgers-Flatiron Postdoctoral Fellow)**,
Present Center for Computational Astrophysics (CCA), Flatiron Institute, NY, USA
- Sep. 2023 – **Postdoctoral Researcher (Joint Rutgers-Flatiron Postdoctoral Fellow)**,
Present Dept. of Physics and Astronomy, Rutgers University, NJ, USA

Education & Research Experience

- Oct. 2019 – **Doctor of Philosophy, Astronomy & Astrophysics**,
Sep 2023 Research School of Astronomy & Astrophysics, Australian National University,
Canberra, Australia
Thesis title: [The interplay of star formation and radiation feedback in the ISM](#)
Awarded the [Charlene Heisler Prize](#) by the Astronomical Society of Australia
Advisors: A/Prof. Christoph Federrath & Prof. Mark Krumholz
- Aug. 2014 – **Integrated Bachelor and Master of Technology (Hons.)**, *Engineering Physics*,
May 2019 Department of Physics, Indian Institute of Technology (IIT),
Varanasi, India
Thesis title: [HI Fragmentation in the Turbulent Interstellar Medium](#)
Advisor: A/Prof. Prasun Dutta
- May 2018 – **Summer Research Scholar**,
August 2019 University of Tübingen,
Tübingen, Germany
Project title: [Impact of ionising radiation on the turbulent ISM](#)
Advisor: Dr. Rolf Kuiper
- May 2017 – **MITACS Globalink Summer Internship**,
August 2017 University of Western Ontario,
London, Canada
Project title: Numerical Magneto-hydrodynamic simulations with the Athena code
Advisor: Prof. Shantanu Basu

Honors & Grants

- July 2024 **Charlene Heisler Prize**,
Awarded by the Astronomical Society of Australia for the most outstanding PhD thesis accepted by an Australian university
- Feb 2023 Joint 4-year Rutgers-Flatiron Postdoctoral Fellowship
- Feb 2023 CCAPP Postdoctoral Fellowship (declined)
- Feb 2023 CITA National Fellowship (declined)
- Oct 2021 ANU Olin J Eggen Research Award (\$2500 AUD),
Awarded to a PhD student for excellence in research

Oct 2022 Vice-Chancellors Travel Grant (\$1500 AUD)
 March 2022 RSAA Travel Grant (\$7000 AUD)
 Oct 2019 ANU International PhD Scholarship (\$29000 AUD p.a.)
 Oct 2019 ANU RSAA Supplementary Research Scholarship (\$ 2500 AUD p.a.)
 August 2018 Graduate Aptitude Test in Engineering (GATE) Postgraduate Scholarship
 May 2018 Eberhard Karls Universität Tübingen Summer Research Stipend
 May 2017 Mitacs Globalink Research Internship Award
 April 2013 Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship

Students Mentored

2 PhD students + 3 undergraduate students

Jan 2025 – **Thummim Mekuria, CCA Predoc Project (PhD student at UC Santa Cruz),**
 May 2025 *Project title: The Feedback-Matter Interplay in High-z Compact Starbursts*
 Jan 2025 – **Avi Chen, PhD student, Rutgers University (co-supervising),**
 present *Project title: [Role of Radiative Feedback in Population III Star Formation](#)*
 Jan 2025 – **Frances Mathias, Undergraduate, Rutgers University,**
 present *Project title: Stochasticity in Stellar Population Synthesis*
 March 2024 – **Dhaval Sai Srinivas, BS-MS student at IISER-Kolkata,**
 present *Project title: The escape fraction of ionising photons in magnetized turbulent media*
 Jan – March 2022 **Hank Hua, undergraduate student at ANU (co-supervised),**
 2022 *Project title: Effects of spiral arms on the Two Point Correlation Function*

Teaching Experience

Sep 2025 **Guest Lecture, Physics 610 – Interstellar Matter,**
Guest lectures on ISM physics for graduate students
 Nov 2024 **Guest Lecture, Physics 345 – Introduction to Computational Astrophysics,**
Guest lecture on parallel computing for graduate students
 Oct 2022 **Guest Lecture, ASTR 4012 – Astrophysical Gas Dynamics,**
Guest Lecture on The Physics of Radiation Hydrodynamics for graduate students
 Feb 2022 – **Teaching Assistant, ASTR 1003 – Astronomy & Space,**
 present *Respond to undergraduate student queries on the course forum and assist in tutorials*
 Aug – Dec 2018 **Teaching Assistant, PHY 1002 – Introduction to Engineering Electromag-**
netics,
Duties involved one tutorial session per week and assistance in preparing tutorials for this undergraduate course
 Jan – May 2018 **Teaching Assistant, PHY 304 – Computational Physics,**
 2018 *Duties involved one laboratory session per week, assisting students in implementing numerical methods and recipes taught in theory classes.*
 March 2022 – **Tutor, Canberra Tutors – High School Mathematics and Physics,**
 present *Conduct private tutorial sessions for grade 11 & 12 students for their physics and mathematics coursework..*

Synergistic Activities

April 2026 **Expert reviewer: National Science Foundation (NSF) Grants**

- July 2021 – **Reviewer: ApJ, ApJL, MNRAS, J. Comput. Phys., JOAA, Open Journal of Astrophysics**
 present
- Sep 2024 – **STEM professional volunteer: Letters to a Pre-Scientist,**
 May 2025 *Program to exchange letters with middle school students to broaden exposure to the day-to-day of STEM researchers.*
- Feb 2022 – **Chair, Seminars Committee, RSAA,**
 present *Heading the seminars team; responsible for organising, hosting and inviting speakers for colloquia*
- Sept 2020 – **Member of Seminars Committee, RSAA,**
 Feb 2022 *Hosted >30 external colloquia and seminars at RSAA*

Proposals

- April 2026 **Co-Investigator**, JWST-GO-10202 : MIRAGE: MIRI Investigation of Red and Ancient Galaxies in the Early Universe
- July 2025 **Co-Investigator**, ALMA Cycle 12 Proposal 2025.1.00874.S: Feeding and Feedback: Connecting Molecular Gas to the Young Stellar Clusters of M83

Seminars and Conference Talks

Invited and Prize Talks

- May 2026 **Astrophysics Seminar: American Museum of Natural History (AMNH)**
- Jan 2026 **The growth of galaxies in the early Universe XI, Sesto, Italy**
- Dec 2025 **Theoretical Astrophysics Seminar: UC Berkeley**
- July 2025 **Review Talk at the Numerical Recipes Lorentz Center, Leiden University**
- June 2025 **Seminar: Raman Research Institute (RRI), Bengaluru**
- March 2024 **ISM Salon, CCA Flatiron Institute**
- Aug 2024 **Cosmic Dawn Revealed by JWST, Kavli Institute of Theoretical Physics**
- June 2024 **Charlene Heisler Prize Talk: ASA Annual Meeting, Online**
- March 2024 **Radiation Transport in Astrophysics, Heidelberg**
- July 2023 **Colloquium: Tata Institute of Fundamental Research, Mumbai**
- July 2023 **Seminar: Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune**
- June 2022 **Seminar: Centre de Recherche Astrophysique de Lyon (CRAL)**
- Dec. 2021 **Colloquium: Indian Institute of Astrophysics, Bangalore**

Contributed Talks

- March 2026 **Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration, Tokyo, Japan**
- April 2025 **Escape of Lyman radiation from galactic labyrinths, Crete, Greece**
- April 2024 **STScI Spring Symposium, Baltimore**
- Feb. 2024 **Building Galaxies From Scratch, Vienna**
- Nov. 2022 **CIERA Theory Seminar, Northwestern University**
- Nov. 2022 **CCAPP Seminar, Ohio State University**
- Nov. 2022 **Lars Hernquist Group Meeting, Centre for Astrophysics, Harvard**
- Nov. 2022 **RandoAstro Seminar, Canadian Institute of Theoretical Astrophysics**
- Oct. 2022 **Coffee Seminar, Institute of Advanced Studies**

- Oct. 2022 **Galread Seminar, Princeton University**
- Oct. 2022 **Cosmology Seminar, MPA Garching**
- Aug. 2022 **Contributed Talk - Star Formation in Different Environments, Quy Nhon, Vietnam**
- July 2022 **Contributed Talk - A Holistic View of Stellar Feedback and Galaxy Evolution, Ascona**
- July 2022 **Planet and Star Formation (PSF) Seminar - Max-Planck-Institut for Astronomy, Heidelberg**
- June 2022 **Star Formation Seminar - CEA, Saclay**
- June 2022 **Contributed Talk - NBIA Workshop on Radiation Transfer in Astrophysics, Copenhagen**
- Nov. 2021 **Contributed Talk, IAU Symposium 362 - The Predictive Power of Computational Astrophysics**
- May 2021 **Contributed Talk, ISM 2021, Beirut - Structure, Characteristic Scales, and Star Formation**
- Feb. 2021 **Contributed Talk, Australian National Insitute for Theoretical Astrophysics science workshop**
- Sept. 2020 **Contributed Talk, Annual meeting of the Astronomische Gesellschaft, Germany**
- March 2020 **Invited Talk, Modelling High-Mass Stellar Feedback Workshop, Tübingen, Germany**
- Jan. 2020 **Australian National Insitute for Theoretical Astrophysics science workshop**

Selected Publications

28 refereed publications: 9 first-author, 5 second-author, 14 collaborator ([ADS Listing](#))

1. [Lyman-alpha Radiation Pressure in Dense Star Clusters: Implications for Star Formation and Winds at Cosmic Dawn](#)
Menon, S., Smith, A. [arXiv e-prints, 2026, arXiv:2605.13982](#)
2. [Bursts of Star Formation and Radiation-driven Outflows Produce Efficient LyC Leakage from Dense Compact Star Clusters](#) [35 Citations]
Menon, S., Burkhardt, B., Somerville, R., Thompson, T., Sternberg, A. [ApJ, 2025, 987, 12](#)
3. [The Interplay between the Initial Mass Function and Star Formation Efficiency through Radiative Feedback at High Stellar Surface Densities](#) [45 Citations]
Menon, S., Lancaster, L., Burkhardt, B., Somerville, R., Dekel, A., Krumholz, M. [ApJ, 2024, 967, L28](#)
4. [Outflows driven by direct and reprocessed radiation pressure in massive star clusters](#) [54 Citations]
Menon, S., Federrath, C., Krumholz, M. [MNRAS, 2023, 521, 5160](#)
5. [Infrared radiation feedback does not regulate star cluster formation](#) [28 Citations]
Menon, S., Federrath, C., Krumholz, M. [MNRAS, 2022, 517, 1313](#)
6. [VETTAM: a scheme for radiation hydrodynamics with adaptive mesh refinement using the variable Eddington tensor method](#) [25 Citations]
Menon, S., Federrath, C., Krumholz, M., Kuiper, R., Wibking, B., Jung, M. [MNRAS, 2022, 512, 401](#)
7. [The dependence of the hierarchical distribution of star clusters on galactic environment](#) [21 Citations]
Menon, S., Grasha, K., Elmegreen, B., Federrath, C., Krumholz, M., Calzetti, D., et al. [MNRAS, 2021, 507, 5542](#)
8. [On the compressive nature of turbulence driven by ionizing feedback in the pillars of the Carina Nebula](#) [38 Citations]

- Menon, S.**, Federrath, C., Klaassen, P., Kuiper, R., Reiter, M. *MNRAS*, 2021, 500, 1721
9. [On the turbulence driving mode of expanding H II regions \[33 Citations\]](#)
Menon, S., Federrath, C., Kuiper, R. *MNRAS*, 2020, 493, 4643
 10. [Blue Monsters and Dusty Descendants: Reconciling UV and IR Emission from Galaxies from \$z\$ 7, up to \$z = 14\$](#)
 Sommovigo, L., Lancaster, L., **Menon, S.**, O'Leary, J., Somerville, R., Bryan, G. *ApJ*, 2026, 1003, 170
 11. [Radiation magnetohydrodynamics simulations of Population III star formation during the epoch of reionization \[3 Citations\]](#)
 van Veenen, L., Sharda, P., Viti, S., **Menon, S.** *Astronomy and Astrophysics*, 2026, 708, A61
 12. [Radiative Feedback in Population III Protostellar Growth: The Impact of HI Shielding](#)
 Chen, A., **Menon, S.**, Burkhardt, B., Sharda, P., Williams, C., Naoz, S., et al. *arXiv e-prints*, 2026, [arXiv:2604.00197](#)
 13. [Density-modulated star formation efficiency: implications for the observed abundance of ultraviolet luminous galaxies at \$z \geq 10\$ \[54 Citations\]](#)
 Somerville, R., Yung, L., Lancaster, L., **Menon, S.**, Sommovigo, L., Finkelstein, S. *MNRAS*, 2025, 544, 3774
 14. [Planets and planetesimals at cosmic dawn: vortices as planetary nurseries \[7 Citations\]](#)
 Eriksson, L., **Menon, S.**, Carrera, D., Lyra, W., Burkhardt, B. *MNRAS*, 2025, 542, 641
 15. [The Coevolution of Stellar Wind-blown Bubbles and Photoionized Gas. I. Physical Principles and a Semianalytic Model \[7 Citations\]](#)
 Lancaster, L., Kim, J., Bryan, G., **Menon, S.**, Ostriker, E., Kim, C. *ApJ*, 2025, 989, 42
 16. [Magnetic fields limit the mass of Population III stars even before the onset of protostellar radiation feedback \[10 Citations\]](#)
 Sharda, P., **Menon, S.**, Gerasimov, R., Bromm, V., Burkhardt, B., Haemmerlé, L., et al. *MNRAS*, 2025, 541, L1
 17. [Population III star formation in the presence of turbulence, magnetic fields, and ionizing radiation feedback \[25 Citations\]](#)
 Sharda, P., **Menon, S.** *MNRAS*, 2025, 540, 1745
 18. [Tracing hierarchical star formation out to kiloparsec scales in nearby spiral galaxies with UVIT \[6 Citations\]](#)
 Shashank, G., Subramanian, S., Muraleedharan, S., **Menon, S.**, Mondal, C., Krishna, S., et al. *Astronomy and Astrophysics*, 2025, 693, A188
 19. [First extragalactic measurement of the turbulence driving parameter: ALMA observations of the star-forming region N159E in the Large Magellanic Cloud \[25 Citations\]](#)
 Sharda, P., **Menon, S.**, Federrath, C., Krumholz, M., Beattie, J., Jameson, K., et al. *MNRAS*, 2022, 509, 2180

Relevant Skills

Languages:, English, Malayalam, Tamil, Hindi, Sanskrit

Programming Experience:, C, C++, Fortran, Python, Mathematica, Bash

Codes/Softwares:, FLASH, PLUTO, PETSc, Hypre, Athena, yt, CUDA

Voluntary Training:, Mental Health First Aid, Understanding and Responding to Sexual Violence